Learning Summary Statistics for the Solar Dynamo

Explicit Noise-Conditional Autoencoders (ENCA) on a stochastic delayed dynamo model

Dipartimento di Fisica e Astronomia Galileo Galilei

Matteo Lovato
ID 2104269

Alberto Casellato
ID 2139206

Advanced Topics in Physics of Data

Academic Year 2025/2026

Master's degree in Physics of Data

Why learn summary statistics?

UNIVERSITÀ DEGLI STUDI DI PADOVA

The problem. The dynamo model is stochastic: its likelihood cannot be computed. Inference is done by simulation (ABC): simulate, compare with the data, accept or reject.

The comparison uses summary statistics $\boldsymbol{s}(\boldsymbol{x})$: a few numbers per time series. They must keep the information about the parameters and ignore the randomness.

In the inference paper (Ulzega et al. 2025): 20 hand-picked FFT components. Reasonable, but arbitrary.

Our goal: learn the summary statistics from simulations, with a neural network (ENCA, Albert et al. 2022). Only the simulator is needed, no real data.

1 - Draw parameters
$\theta \sim prior$

⬇

2 – Simulate
$x = M(\theta, \varepsilon)$ (Julia SDDE)

⬇

3 – Compress
$s = s(x)$ ← the object we learn

⬇

4 – Compare
$\|s(x) - s(x_{obs})\| < \varepsilon_{tol}$ → posterior

Plan: 1 parameter → interpret the latent space → 2 and 4 parameters → chaos → Fourier → full prior range

The solar dynamo model

$$(d/dt + 1/\tau)^2 \, B(t) = -(Nd/\tau^2) \, f(B(t-T)) \, B(t-T) + (\sigma \, Bmax/\tau\text{^}(3/2)) \, \eta(t)$$

x(t) = B²(t) is the proxy of the sunspot number (Wilmot-Smith et al. 2006; Albert et al. 2021)

τ → diffusion time: how fast the field decays; sets the time scale

T → delay: time for the field to travel between the two regions where it is generated

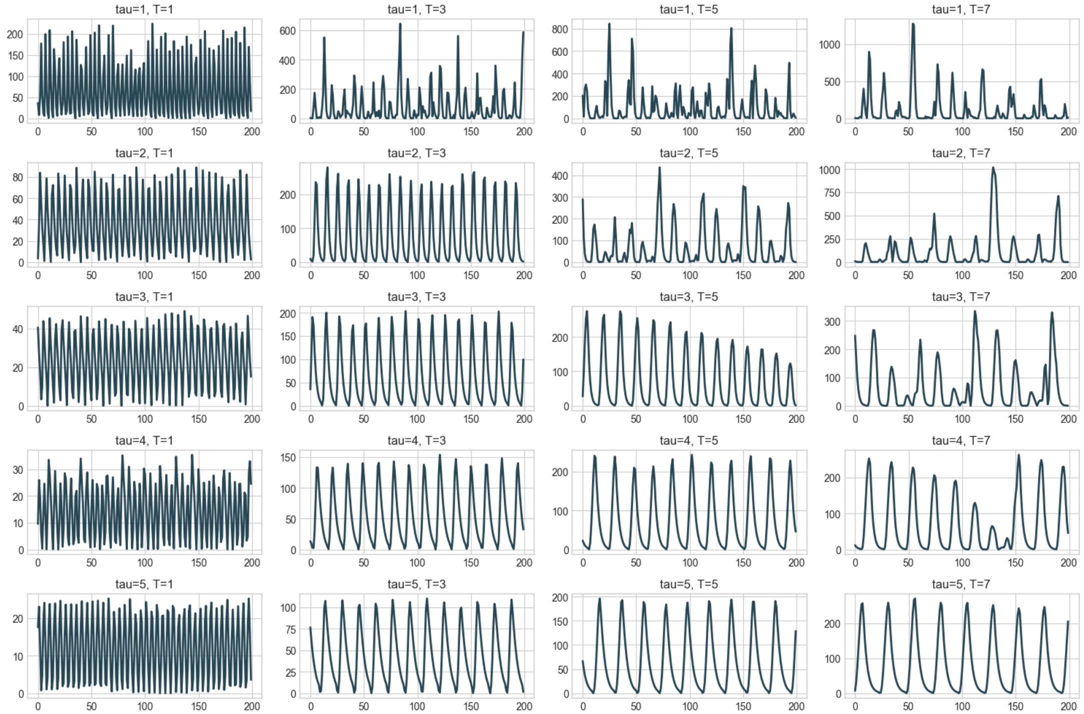
Nd → dynamo number: drive vs dissipation; controls the dynamics

σ → noise amplitude (turbulent convection)

Bmax → sunspot-formation threshold; only rescales the amplitude → kept fixed

Why a delay? The equation contains $B(t-T)$: the field now depends on the field T time units ago, because the two field components are generated in different layers of the Sun. The delay produces cycles, chaotic modulation, and two coexisting modes: weak and strong (bistability, linked to Grand Minima).

Parameter scan. The (τ, T) grid on the right shows the regimes: regular for small T, chaotic for large T. We start from the safe point $\tau = 2.8, T = 3.7$ and extend later.



(τ, T) scan with $Nd = 11, \sigma = 0.05$: regular → chaotic as T / τ grows

Requirements for the summary statistics

Sufficiency Condition

$$I(\Theta, s(X)) \approx I(\Theta, X)$$

The statistics keep (almost) all the information that the series carries about the parameters.

Concentration Condition

$$H(S|\Theta) \approx -\ln N$$

Two runs with the same θ must give similar statistics: the statistics ignore the noise. This is what lets the ABC tolerance shrink.

How many?

$$p + 2$$

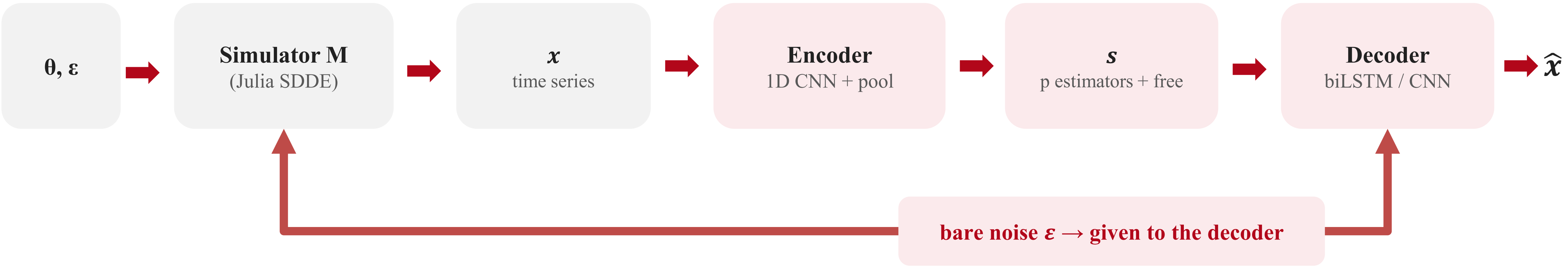
One number per parameter is not enough: runs with the same θ still differ (phase, regime). We add two free dimensions: $n\_summary = n\_params + 2$.

Why it matters here. In this model the same θ can produce different behaviours (bistability).

The statistics must also record which behaviour occurred, and this cannot be stored in the parameter estimators.

We test the choice of n_summary directly with an ablation. [Albert et al. 2022]

The ENCA method



Key idea. The decoder receives the exact noise used by the simulator. Encoding noise is therefore useless: the encoder can only improve the reconstruction by passing parameter information → s becomes a minimal set of summary statistics.

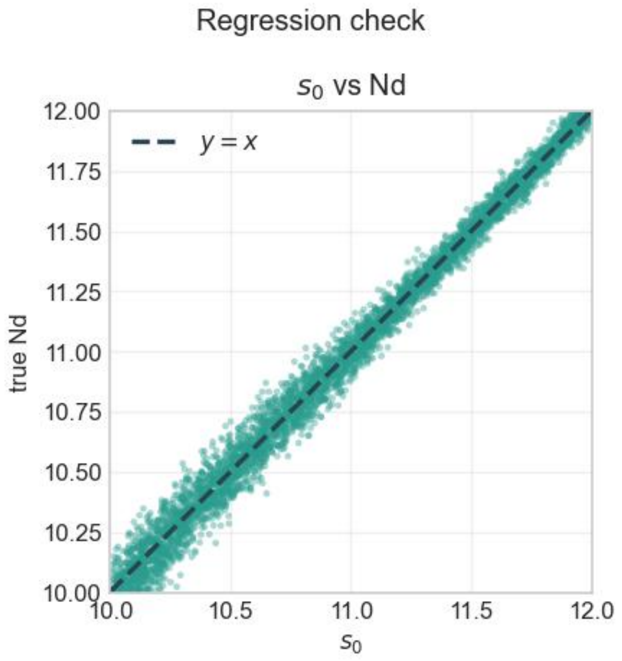
Loss. Reconstruction $MSE(\hat{x}, x) + regressionMSE(s_{0} \ldots {}_{p}, \theta)$. The first p summaries become parameter estimators, checked with s vs θ scatter plots ($y = x$ line). The other dimensions are free.

Networks. Encoder: 1D CNN + global average pooling. Decoder: biLSTM in the time domain, CNN on spectra in Fourier, FNO tested as extension.

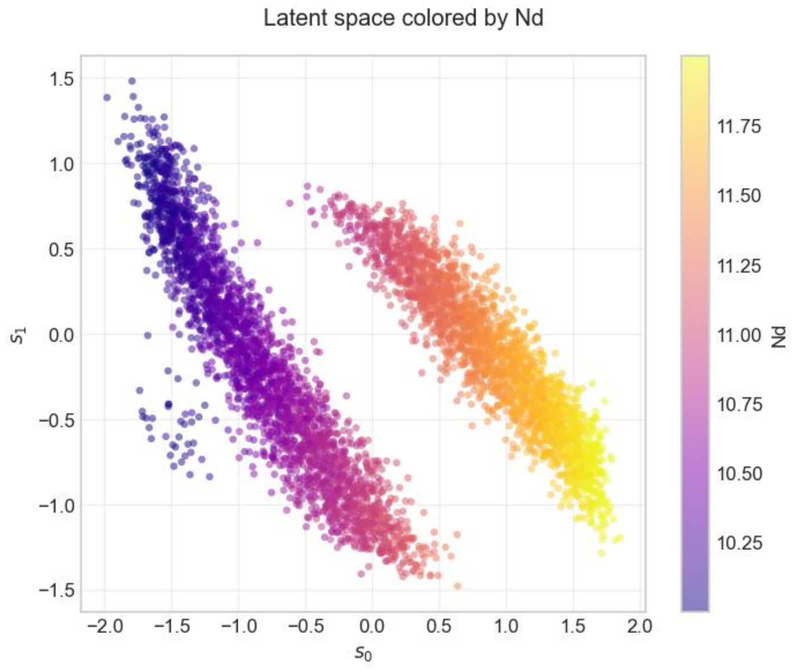
First experiment: one parameter (Nd)

$Nd \sim U(10, 12)$, other parameters fixed; time domain, biLSTM decoder, $n\_summary = 2$, 150k samples.

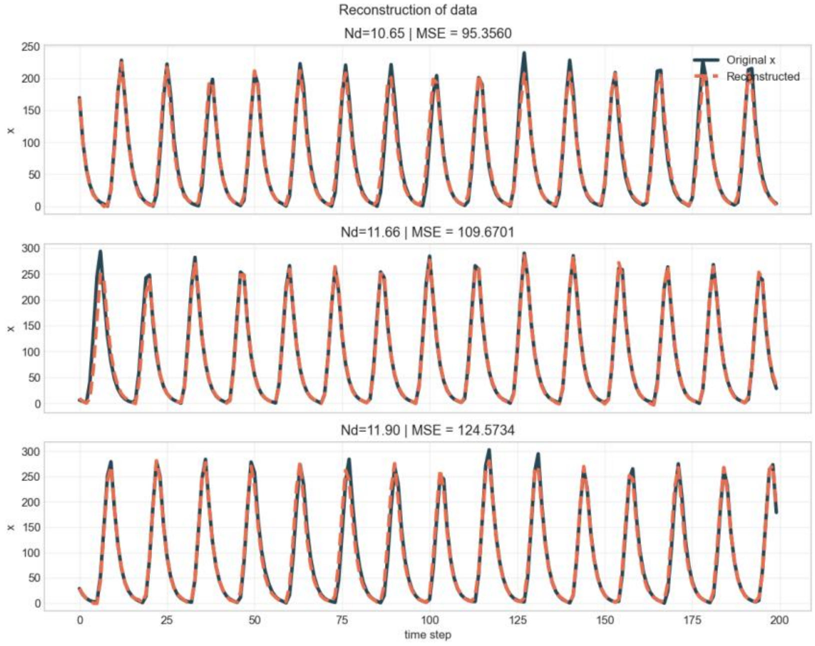
The single summary recovers Nd ($y = x$) but the latent space splits into two clusters.



s_0 vs Nd: on the $y = x$ line



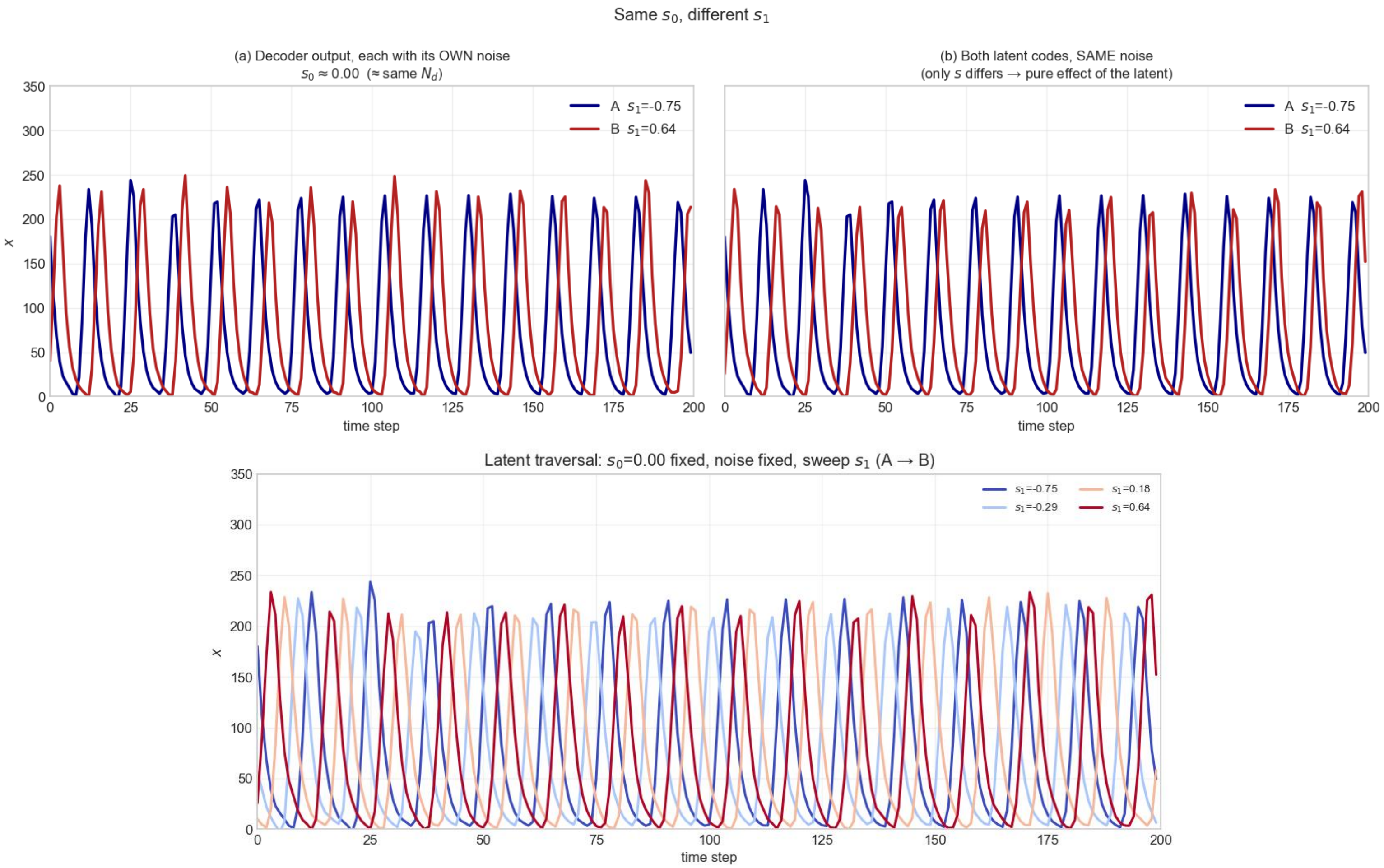
latent space (s_0, s_1) coloured by Nd: two branches, smooth gradient in each



ENCA data reconstruction for 3 Nd values

Interpreting the free dimension: the phase

UNIVERSITÀ DEGLI STUDI DI PADOVA

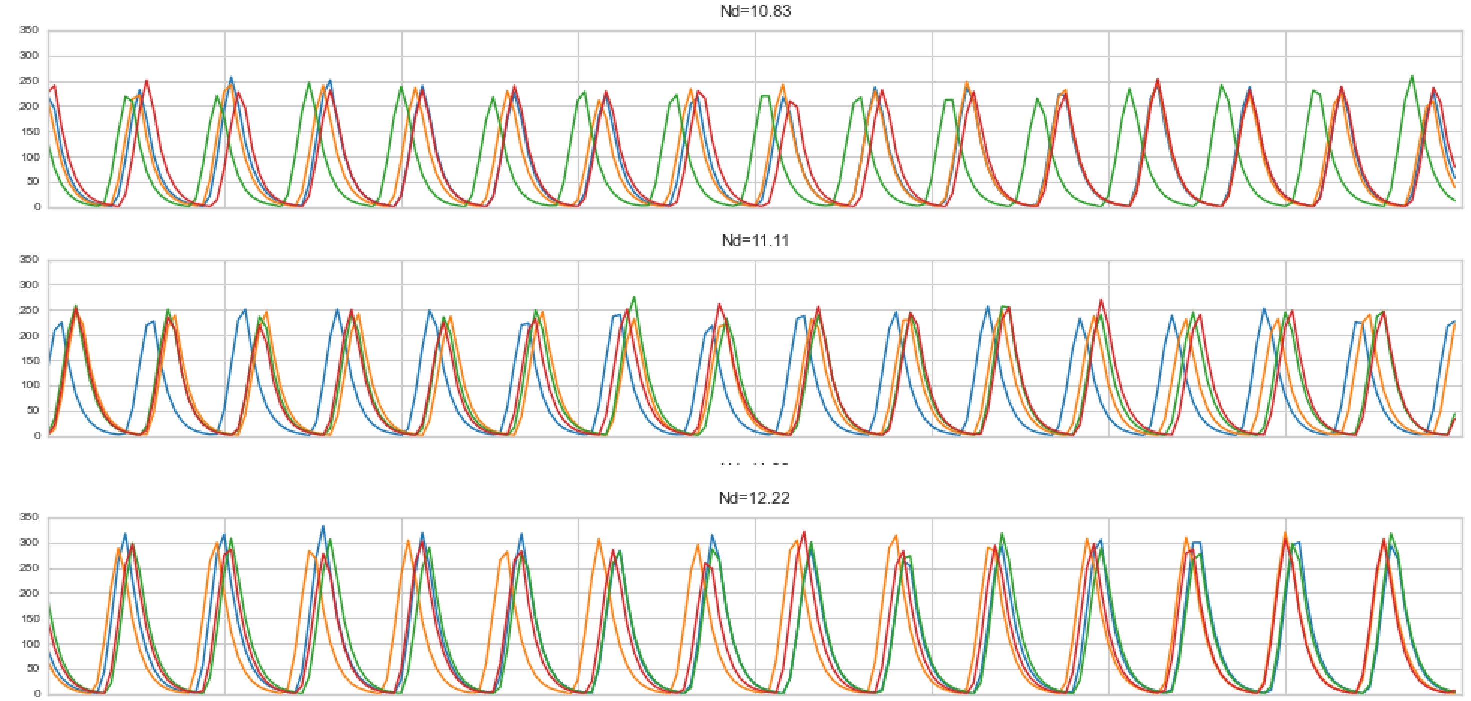


Test. Two runs with the same s_0 (same Nd) but opposite clusters. (a) each decoded with its own noise; (b) both decoded with the same noise → the only difference left is s_1.

Result. Same period, same amplitude, only a time shift. Sweeping s_1 continuously (bottom figure) shifts the phase smoothly.

Conclusion. s_1 stores the phase of the cycle: information about the single run, not about the parameters.

Bistability check



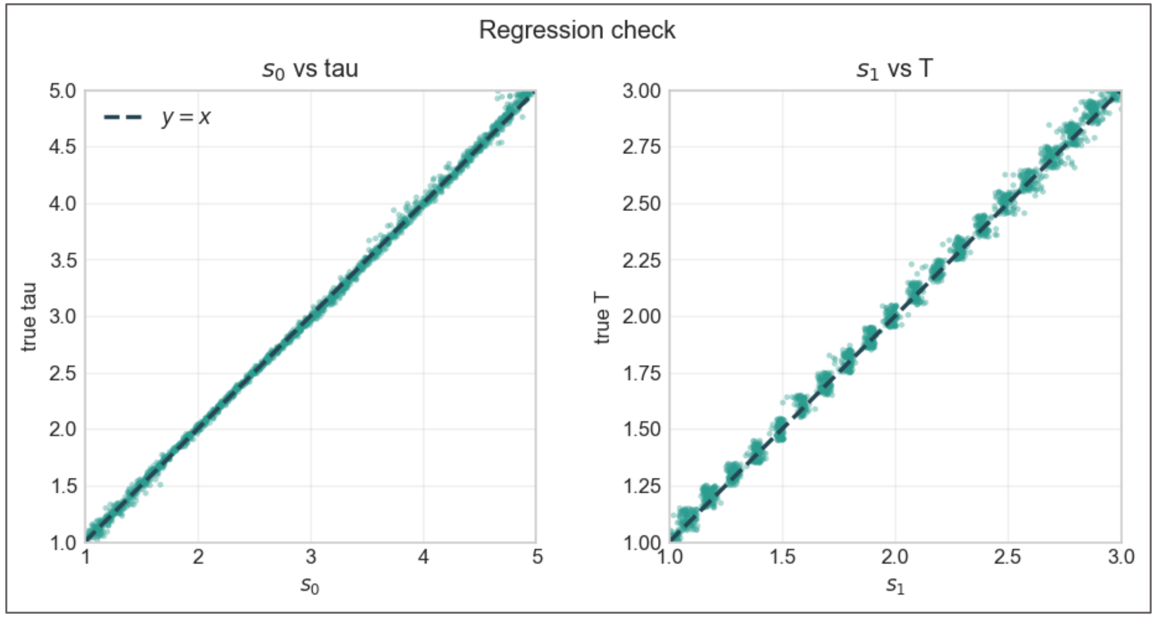
Same Nd, different seeds - top: transition region ($Nd \approx 10.8 - 11.1$), realizations diverge; bottom: $Nd = 12.2$, seed-independent

The test. One value of Nd, same parameters, different random seeds. No neural network involved.

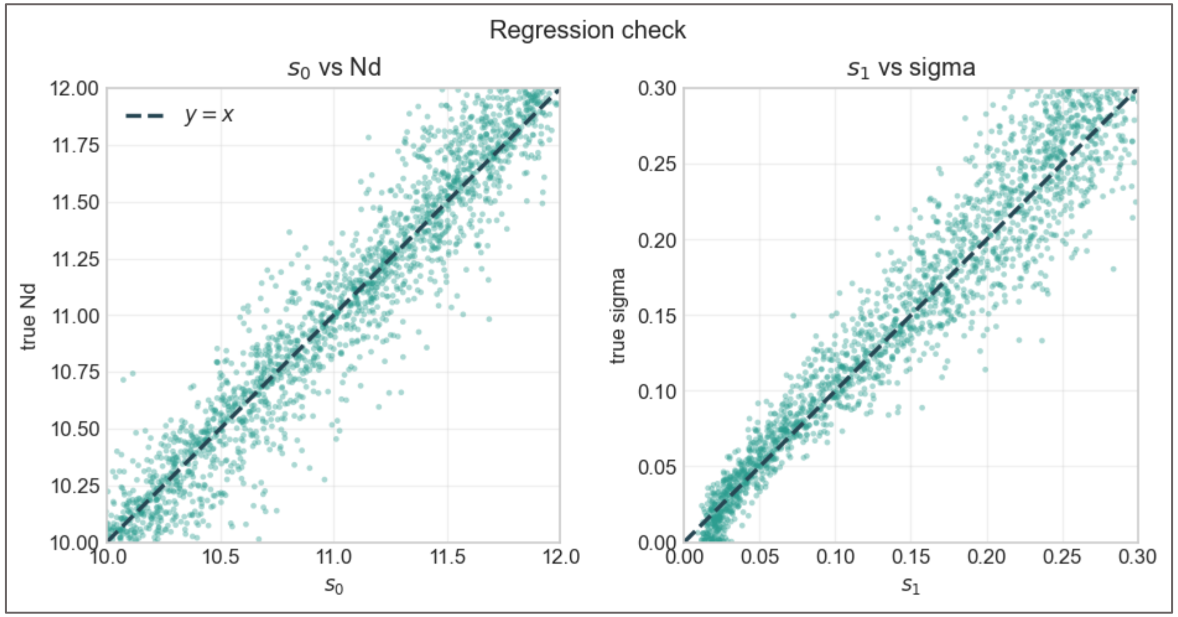
Result. In the middle of the range ($Nd \approx 10.8 - 11.1$) some runs settle on a strong regular cycle, others on a weaker irregular one. At the edges of the range all runs look alike.

Meaning. The two clusters reflect a real property of the model: **bistability**. A summary must also record which behaviour occurred, this is the role of the free dimension.

Two varying parameters



(τ, T) vary - note the staircase in T



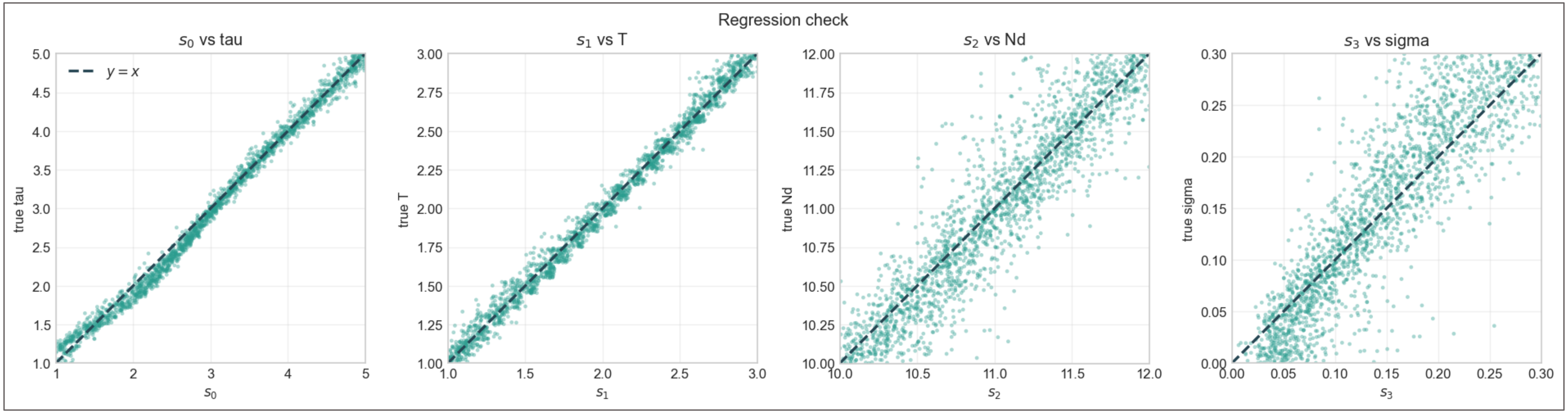
(Nd, σ) vary - σ is intrinsically noisier

The staircase in T: the simulator rounds T to its 0.1 integration grid, so T is effectively discrete; the network learns the steps.

σ is harder: it is a variance, poorly determined by a single series; accuracy drops as the noise grows.

Four varying parameters

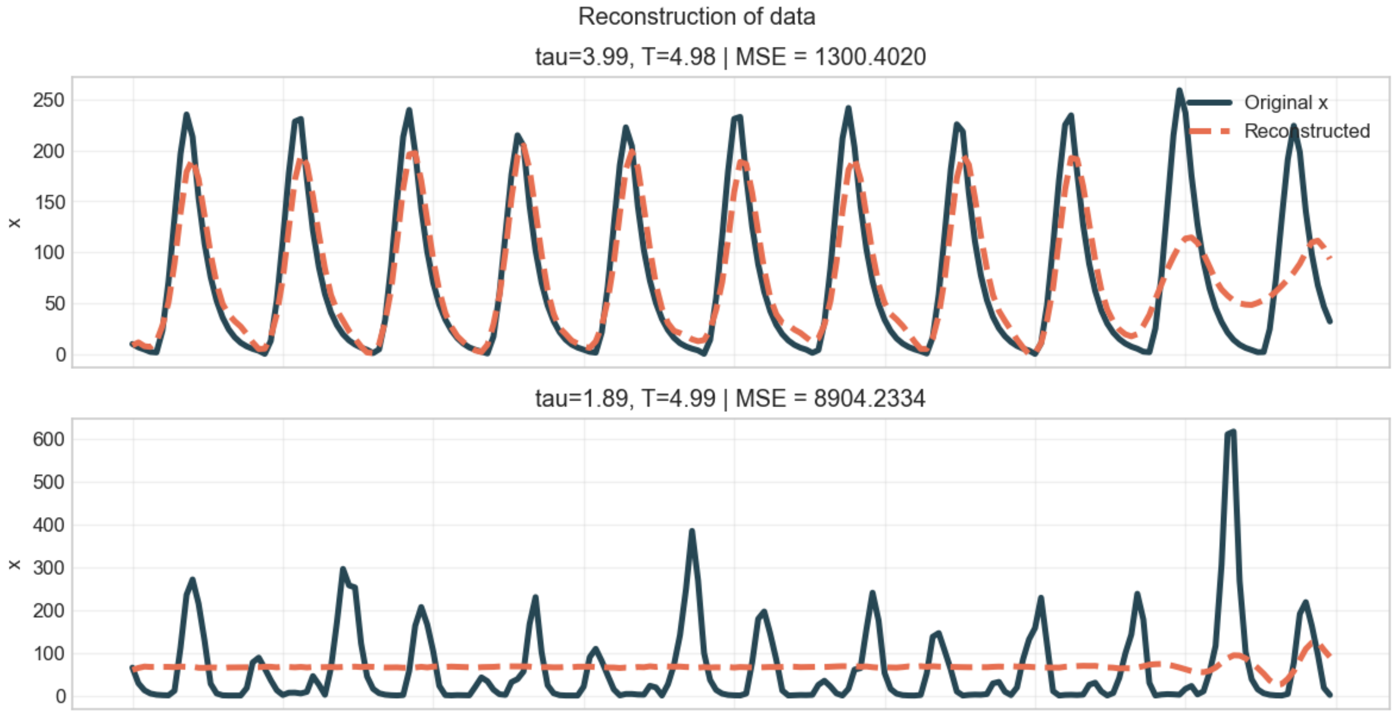
Restricted priors away from chaos; time domain, $n\_summary = 6$ (Bmax fixed).



τ and T are recovered well: the period of the cycle is a clear fingerprint.

Nd and σ are harder: within the restricted prior, different (Nd, σ) pairs give almost identical series, a partial identifiability problem.

Extending the prior into chaos



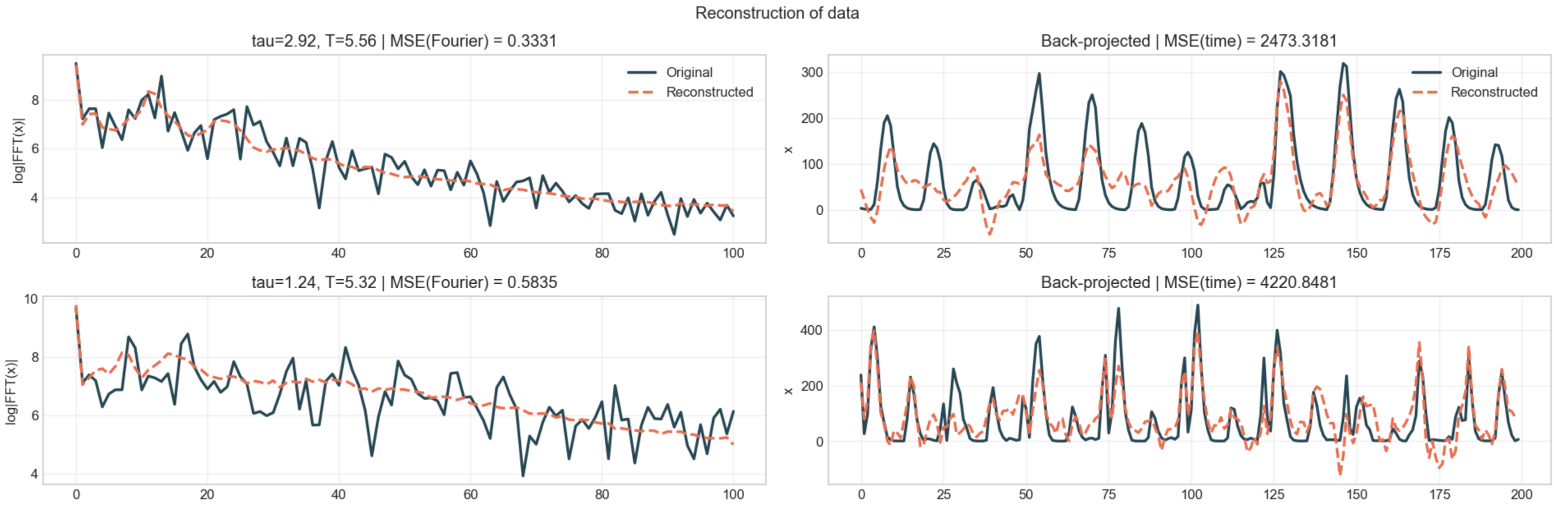
T ~ U(1,7), time domain: fine on regular samples (top), hopeless on chaotic ones (bottom)

Setup. Same (τ, T) experiment, but the prior of T extends to $U(1,7)$: many runs are now chaotic.

Result. The reconstruction breaks: a chaotic run cannot be matched point by point (bottom sample: $MSE \approx 8900$). τ and T are still regressed well, the period survives chaos, but the reconstruction is the mechanism that makes the statistics complete, and it is gone.

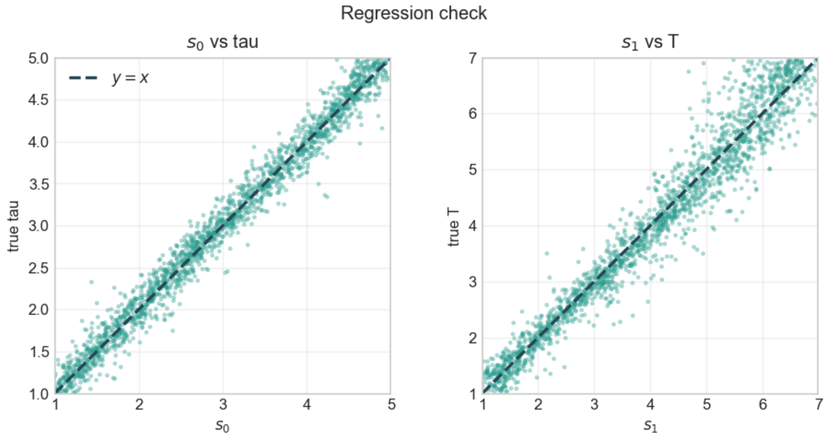
Conclusion. Fixing it needs a better representation of the data, not a bigger network → **Fourier space**.

Moving to Fourier space



Setup. Encoder and decoder (CNN) work on log-magnitude spectra; the loss is computed in Fourier space.

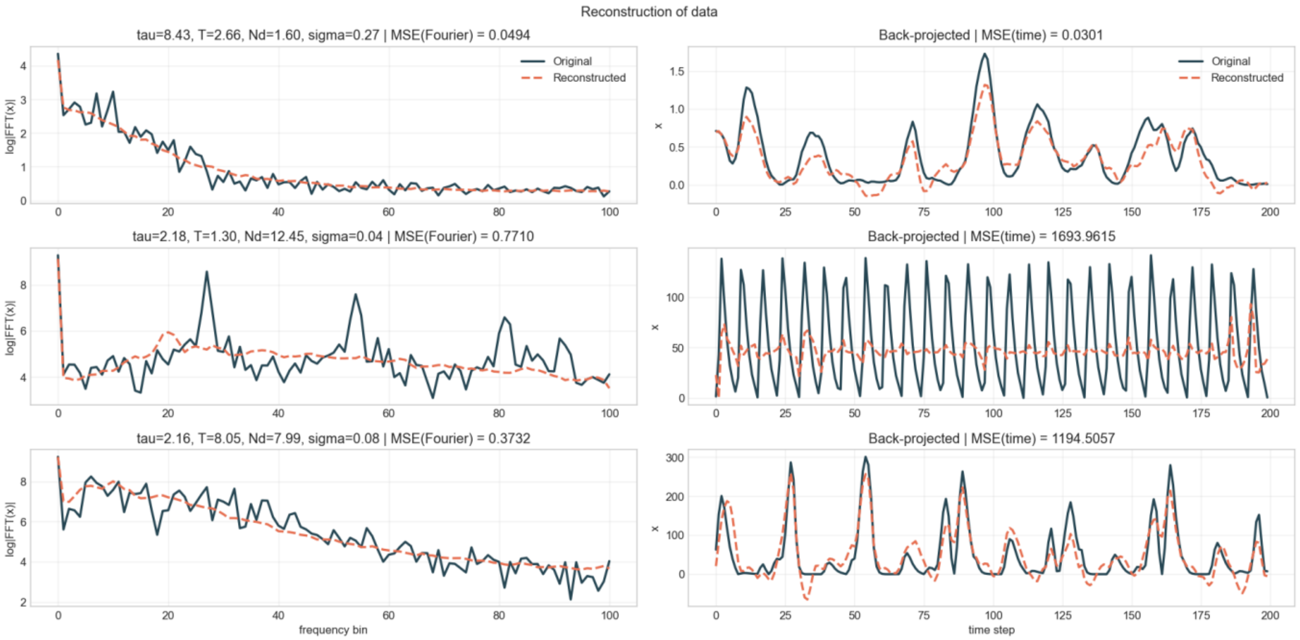
Why it helps. The parameter information sits in the dominant peak; chaos becomes a broadband background and no longer dominates the loss.



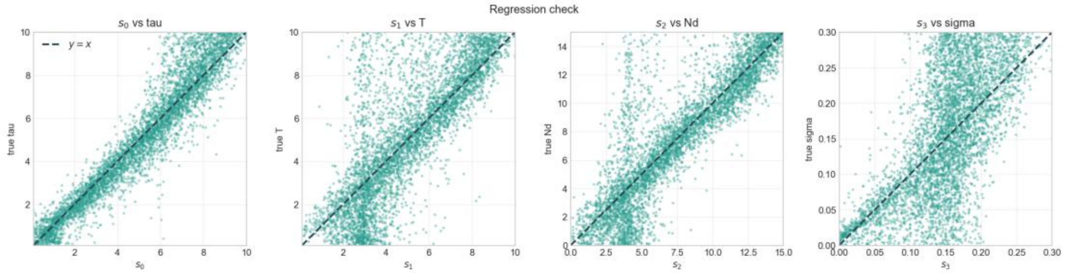
Same $T \sim U(1, 7)$ experiment: spectra well reconstructed, dominant cycle recovered, and the τ and T regression is clean over the whole prior, chaos included.

Main result: full prior range

$\boldsymbol{\tau, T \sim U(0.1, 10), \quad Nd \sim U(0, 15), \quad \sigma \sim U(0, 0.3)}$ **; 150k simulations, Fourier,** $\boldsymbol{n\_summary = 6.}$



log-magnitude spectrum (left) and back-projected series (right) — three test samples across regimes

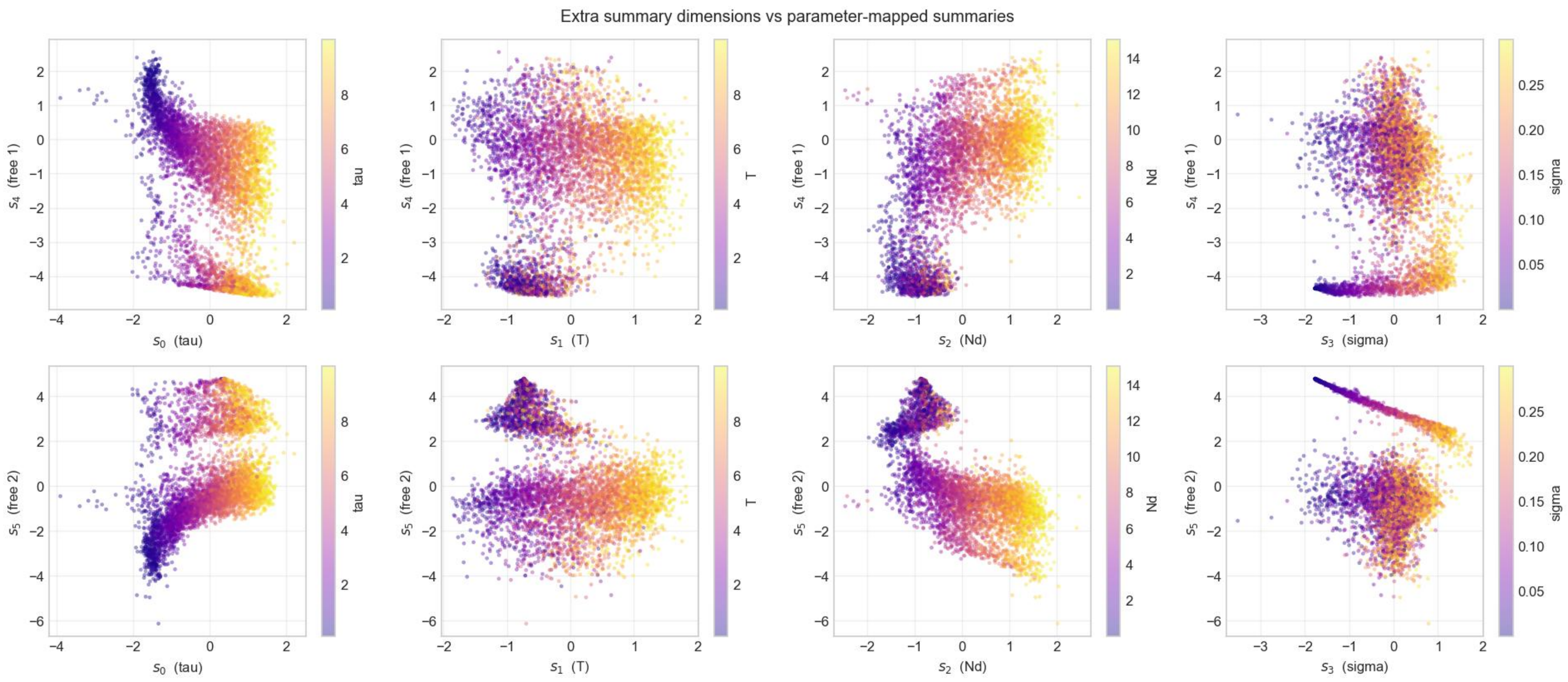


Vertical bands come from the dead-dynamo region (Nd below critical: B decays, little signal to regress on).

Reconstructions. Spectra are reconstructed well across all regimes, and the back-projected series recover the dominant cycle, chaotic runs included. The CNN smooths the sharpest spectral peaks, so fast regular cycles lose some amplitude after back-projection.

The regression works over the entire prior of the inference paper where the time-domain attempts had failed.

Main result: the extra summaries

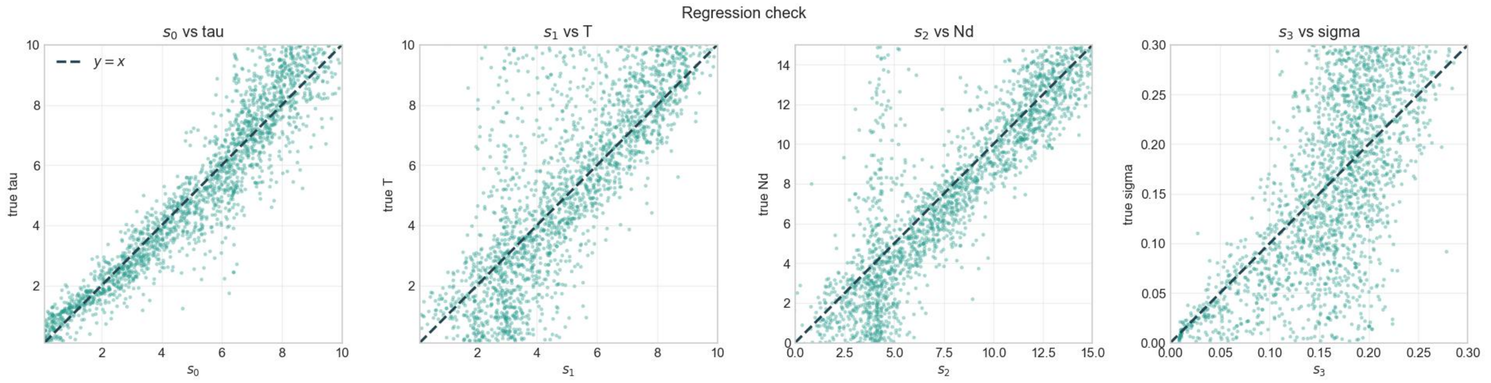


The free dimensions s_4 and s_5 separate into two manifolds: the regime information (**bistability**) is still there, now in a 6-dimensional latent space.

Additional tests: ablation and FNO decoder

Ablation: *n_summary* = *n_params*

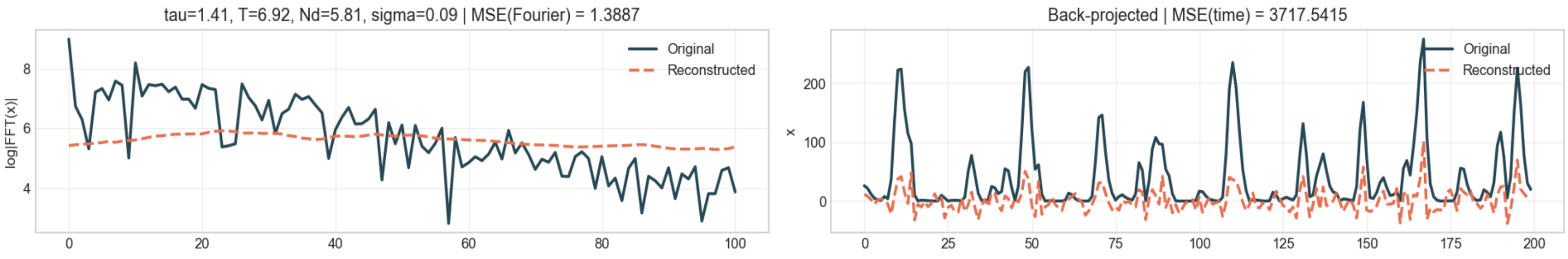
Bottleneck reduced to $4 = p$.
Performance drops, most clearly in the reconstruction; the regression gets somewhat blurrier.



regression with n_summary = 4

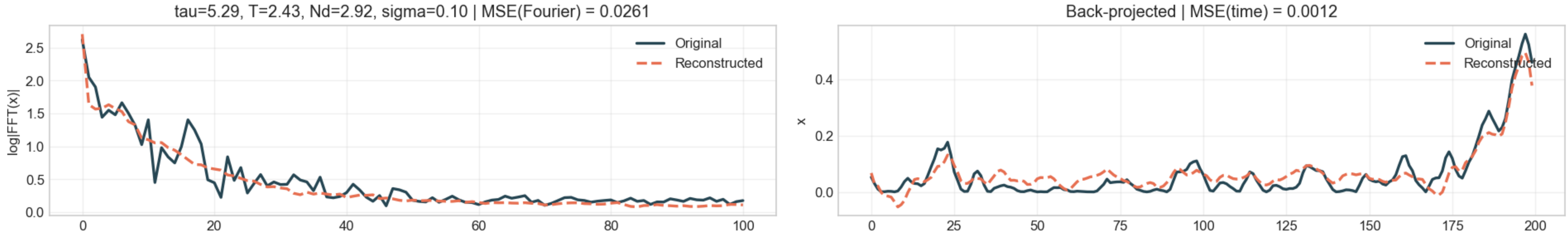
FNO decoder

Fourier Neural Operator as decoder.
Results are not satisfactory despite the tests → left as future work.



FNO spectra

Additional tests: noise in Fourier space



Fourier-noise reconstruction

The noise is transformed like the output.

Better reconstructions, regression unchanged, total loss 0.327 vs 0.334 of the main run.

Conclusions

Università degli Studi di Padova

1 **ENCA works on the solar dynamo.** The learned summaries are interpretable and near-minimal; in the easy regime the estimators sit on the $y = x$ line.

2 **The data representation matters.** The time domain fails in chaos; in Fourier space the full prior of the inference paper becomes tractable.

3 **The free dimensions have physical meaning.** They store the phase and the regime (bistability), confirmed by multi-seed simulations and by the ablation.

4 **Next steps.** Train the FNO decoder properly.
Apply the encoder to the real sunspot record and check whether the data falls inside the prior range.